Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wave has frequency 4 Hz and wavelength 6 m. Find its speed.

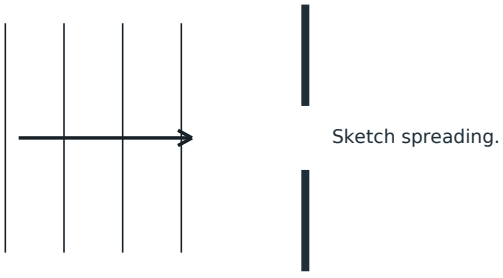
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\lambda = v/f$; compare opening width with λ. Spreading is pronounced when the opening width is comparable to wavelength. Label the opening and wavefronts.

Worked reasoning: A wave with wavelength 1 m meets openings 1 m and 10 m wide. Spreading relative to opening size is more noticeable for the 1 m opening.

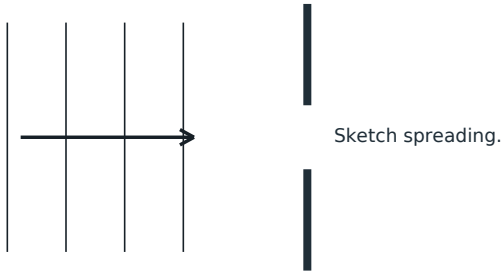
Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.



Marzano check - circle: ① ② ③

03 Read and interpret the model

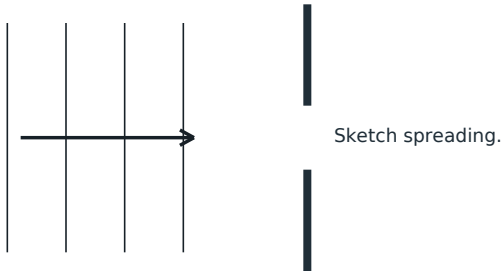
Sketch wavefronts after a narrow opening about one wavelength wide. Compare with a much wider opening.



Marzano check - circle: ① ② ③

04 Apply the physics

A ripple-tank simulation sends 1 m wavelength waves toward two openings: A is 1 m wide and B is 10 m wide. Sketch the emerging wavefronts. Which opening produces more noticeable spreading relative to its width?



Marzano check - circle: ① ② ③

05 Explain your reasoning

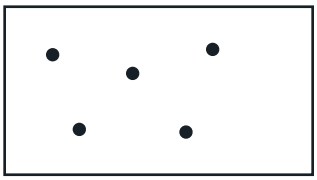
A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Marzano check - circle: 1 2 3

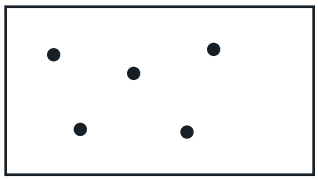
06 Review: connect an earlier idea

At the same kelvin temperature, sample B contains 4 times as many particles of the same ideal gas as A. Compare average energy per particle and total translational energy.
Use particle arrows and labels to show the comparison.

Sample A



Sample B

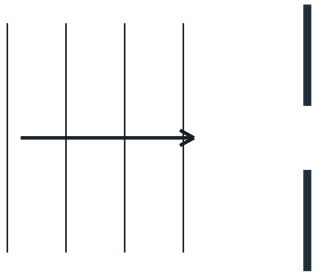


Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

For the same opening, would a longer wavelength generally spread more or less?

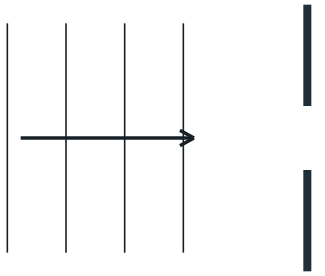


Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep opening width fixed and halve wavelength 0. Predict the change in relative spreading and draw before/after patterns.



Sketch spreading.

Marzano check - circle: 1 2 3